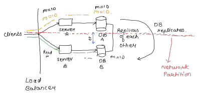
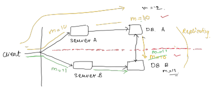
Agenda :
- CAP Theorem
- PACELAC
- Data Consistency.

★ Distributed Systems.



What's happening ?
- clients can hit any server.
- server talks to nearest DB replica.
- DB replicates among them self.



- DB Node A cannot talk to DB node B.
- This is called partition (P).

In case of network partion, the Data will be inconsistent.

★ CAP Theorem – whatever that can be proved to be true.
└ Brewer's theorem.

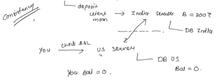
In Distributed System,
Only two of them can be achieved.
- Consistency
- Availability
- Partition tolerence

i.e. CA / AP / CP

★ Consistency – Every read receives the most recent write.

- m=15 on node A,
the Node B return m=15.

Ex1 :
you mom deposit 200 ₹ from India to HDFC AC



★ Availability
- Every req (read/write) receives a response. Even if the data is inconsistent or stale.
- The system should be always available.

★ Partition tolerance.
└ The system should continue to operate despite network failure (Partitions) between the node.

└ if Node A ✕ Node B due to network issue,
the system should work.

Ex : Restaurent – 24x7 Mumbai
- Chef is sick
- Ingredients are missing.

- The restaurent still serves the food.
  - └ quality may drop
  - └ menu might be limited
  - └ Everyone will get the food.

} Availability.

Sync the sales data

Restaurent – 24x7 Bangalore.
- same menu.
- order Book.
- Sync of sales data not available b/w mum & BLR

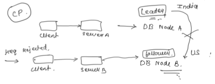
Partition

Mum Rest → available.
BLR Rest → available.
} Both the system should be individualy able to operate.

Even if the Branches can not talk, they still keep operation.

CAP cont.

CP – consistency + Partition tolerence
AP – Availability + Partion tolerence
CA – consistency + Availability.



Consitensy over availability.
→ If DB is unsure about the data, Request will rejected.

- No stale Date.
- get error (req might get rejected).

Bank : – If ATM is not able to decide the BAL is correct or not.
- service unavilable.
- Better than showing wrong Bal.

AP

client → server A → DB (A) writes allowed
client → server B → DB (B) writed allowed.

Both should continue working indenpendtly.
- Availabililty over consistency.
- every req will be served – System never goes down.

Ex : Instagram.
- Likes count.

Canandra
Dynamo DB.

✕ ⇒ CA consistency + Availability ✕ Conflh in distributes CAP

CA is possible only when there is no Partition

If Partition ⇓ OR Else ↑ OR
★ PACELC Theorem.

If there is a partition (P),
the system must choose between
✓ Availability (A) OR → FB, Insta
✓ Consistency (C) → Banking.

else (Partion is no partition)
it must choose latency (L) ✓ – INSTA.
OR
consistency. ✓
└ BANK